

Vignesh Sitaraman

PhD Candidate specializing in experimental nuclear physics, detector construction, and digital DAQ systems.

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EDUCATION

- **Ph.D. in Nuclear Astrophysics** — Florida State University, Tallahassee, FL (Present)
- **M.S. in Physics** — Florida State University, Tallahassee, FL, 2020
- **M.Sc. in Physics** — BITS Pilani – Hyderabad Campus, India, 2018 (Thesis: Digital Signal Processing for the Charged Particle Detector Array at INGA)
- **B.E. in Chemical Engineering** — BITS Pilani – Hyderabad Campus, India, 2018 (Thesis: Desalination of Water using Nano-Photonics)

RESEARCH EXPERIENCE

- **Graduate Research Assistant**, John D. Fox Superconducting Linear Accelerator Laboratory, FSU — May 2020 to Present
Supervisor: Prof. Ingo Wiedenhöver. Conceptualized and implemented a novel design upgrade to the internal tracking proportional counter, demonstrating an order-of-magnitude improvement in position resolution. Deployed the upgraded detector in three experimental campaigns measuring the $^{27}\text{Al}(\alpha, p)$, $^{24}\text{Mg}(\alpha, p)$, and $^{17}\text{F}(\alpha, p)$ reactions. Upgraded the digital DAQ system by integrating CAEN 1740/1725 digitizers and QDC/PHA firmware for coincident readout using FSUDAQ. Assisted in setting up digital electronics and trigger logic for the SABRE array with the SE-SPS spectrograph using CoMPASS.
- **Junior Research Fellow**, Tata Institute of Fundamental Research (TIFR), Mumbai — Aug 2018 to Jul 2019
Supervisors: Prof. Rudrajyoti Palit and Prof. S.N. Mishra. Evaluated timing and energy response of $\text{LaBr}_3(\text{Ce})$ detectors for the INGA array; investigated structural properties of neutron-deficient nuclei in the mass-130 region.
- **Project: Desalination of Water using Nano-Photonics**, BITS Pilani – Hyderabad Campus — Jan 2018 to May 2018
Supervisors: Prof. Kannan Ramaswamy and Dr. Vikranth K. Surasani. Simulated a Nano-Photonics Enabled Solar Membrane Distillation (NESMD) plant with a bilayer membrane architecture and evaluated its geographic viability for Hyderabad.
- **Project: Digital Signal Processing for the Indian National Gamma Array**, TIFR, Mumbai — Aug 2017 to Dec 2017
Supervisors: Prof. Rudrajyoti Palit and Dr. Sarmistha Banik. Implemented and compared DSP algorithms to analyze high-volume data from INGA and the Charged Particle Detector Array (CPDA).

PUBLICATIONS

- 2026 The study of the $^{17}\text{F}(\alpha, p)$ reaction — *In Preparation*
- 2026 The Upgraded ANASEN Proportional Counter — *In Preparation*
- 2026 Properties of states in ^{19}Ne important for the $^{18}\text{F}(p, \alpha)^{15}\text{O}$ reaction rate — *arXiv preprint*
- 2026 The ICESPICE demonstrator for particle/ γ - e^- coincidence experiments at Florida State University — *Nuclear Instruments and Methods in Physics Research A*
- 2026 Measurement of near-threshold resonances in ^9B for Big Bang nucleosynthesis — *Physical Review C*
- 2025 Experimental study of ^{53}Cr via the (d, py) reaction — *Physical Review C*
- 2025 Study of α -resonances in ^{11}B via the $^7\text{Li}(^7\text{Li}, t)$ reaction — *Physical Review C*
- 2024 Single-neutron adding on ^{34}S — *The European Physical Journal A*
- 2024 The CeBrA demonstrator for particle-gamma coincidence experiments at the FSU Super-Enge Split-Pole Spectrograph — *Nuclear Instruments and Methods in Physics Research A*

- 2024 Suppressed electric quadrupole collectivity in ^{49}Ti — *Physics Letters B*

TALKS & PRESENTATIONS

- 2025 The study of the $^{17}\text{F}(\alpha, p)$ reaction using the upgraded ANASEN detector — APS DNP Fall Meeting, Chicago, IL
- 2024 The study of the $^{27}\text{Al}(\alpha, p)$ reaction using the upgraded ANASEN detector — APS DNP Fall Meeting, Boston, MA
- 2023 Upgrades to the ANASEN Active Target Detector for (α, p) reactions with radioactive beams — Gordon Research Conference, New London, NH
- 2021 The study of the $^{17}\text{F}(\alpha, p)^{20}\text{Ne}$ reaction rate using the ANASEN Active Target detector — APS DNP Fall Meeting, Boston, MA
- 2018 Neutron Star Workshop — BITS Pilani, Hyderabad, India
- 2018 PlasmaSurf Summer School for Nuclear Fusion and Plasma Physics (ITER/IPFN) — IPFN, Lisbon, Portugal
- 2018 Desalination of Water using Nanophotonics — National Symposium on Emerging Environmental Challenges, Hyderabad, India

TEACHING

- **PHY 2054 (College Physics B):** Undergraduate lab sessions (Electromagnetism, Circuits, Optics), 2020
- **PHY 2053 (College Physics A):** Undergraduate lab sessions (Mechanics, Kinematics, Thermodynamics), 2019–2020
- Conducted pre-lab lectures, graded technical reports, and hosted weekly office hours.

MENTORSHIP & SERVICE

- Mentored junior students and undergraduates in experimental nuclear physics techniques: vacuum systems, digital electronics, detector testing/assembly, and data analysis/simulation fundamentals using C++ and ROOT.
- **Tandem Maintenance Coordinator**, John D. Fox Laboratory — trained incoming graduate students to conduct lab safety and maintenance checks and organized the maintenance schedule.

TECHNICAL SKILLS

- **Detector Instrumentation:** Detector design and fabrication; gas-filled proportional counters, silicon strip detectors, HPGe detectors; high-vacuum environments; analog/digital data acquisition.
- **Data Acquisition (DAQ):** CAEN 1740/1725 digitizers, digital DAQ firmware configuration (PHA/QDC), FSUDAQ, CoMPASS, hardware troubleshooting.
- **Data Analysis & Simulation:** ROOT, Garfield++, Gmsh, Elmer, ParaView, pycatima, SRIM, LISE++, MATLAB, Octave.
- **Programming Languages:** C++, Python, LaTeX.
- **Engineering & Modeling:** SolidWorks, COMSOL.
- **Accelerator Operations:** Operational proficiency with superconducting LINAC and Tandem Van de Graaff accelerators.
- **Languages:** English, Hindi, Tamil, Spanish (B1).

AWARDS & MEMBERSHIPS

- John D. Fox Award, Department of Physics, Florida State University, 2025
- Evelyn and John Baugh Scholarship, Department of Physics, Florida State University, 2024
- Dean's Fellowship, Florida State University, 2019
- Junior Research Fellow, Tata Institute of Fundamental Research, 2018
- Member, American Physical Society, Present

- AAAS Scholarship, Member, American Association for the Advancement of Science, 2020–2021